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FASTPTX HANDS-ON SESSION

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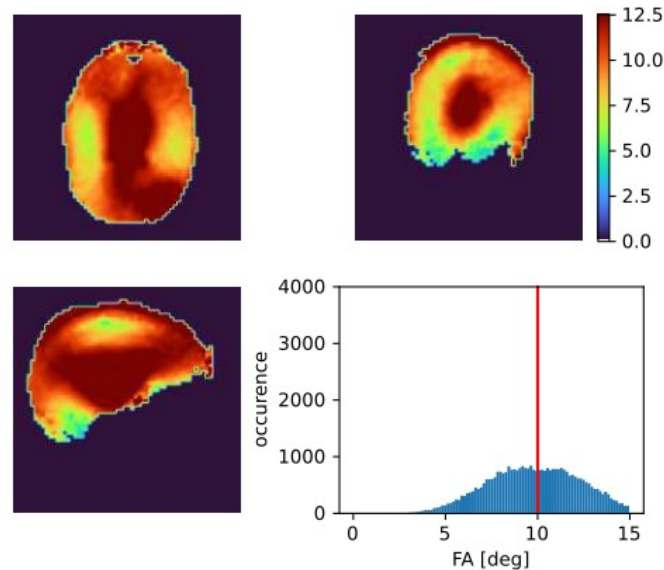
ESMRMB Annual Meeting 2025, Marseille



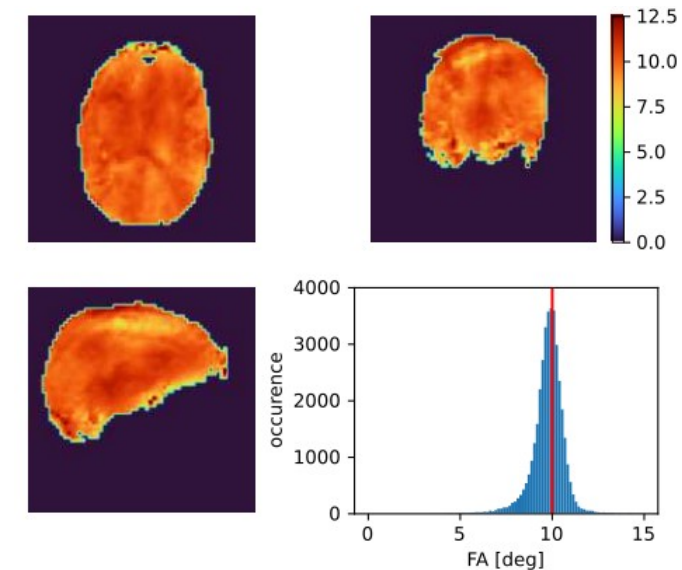
PARALLEL TRANSMISSION IN ULTRAHIGH FIELD MRI

MRI at ultrahigh fields:

- Inhomogeneous B_1^+
- Highly localized tissue heating $\rightarrow$ SAR
- Parallel Transmit Systems: many RF (B_1^+) transmitters



CP-mode excitation FA



pTx excitation FA

Very expensive high-end systems

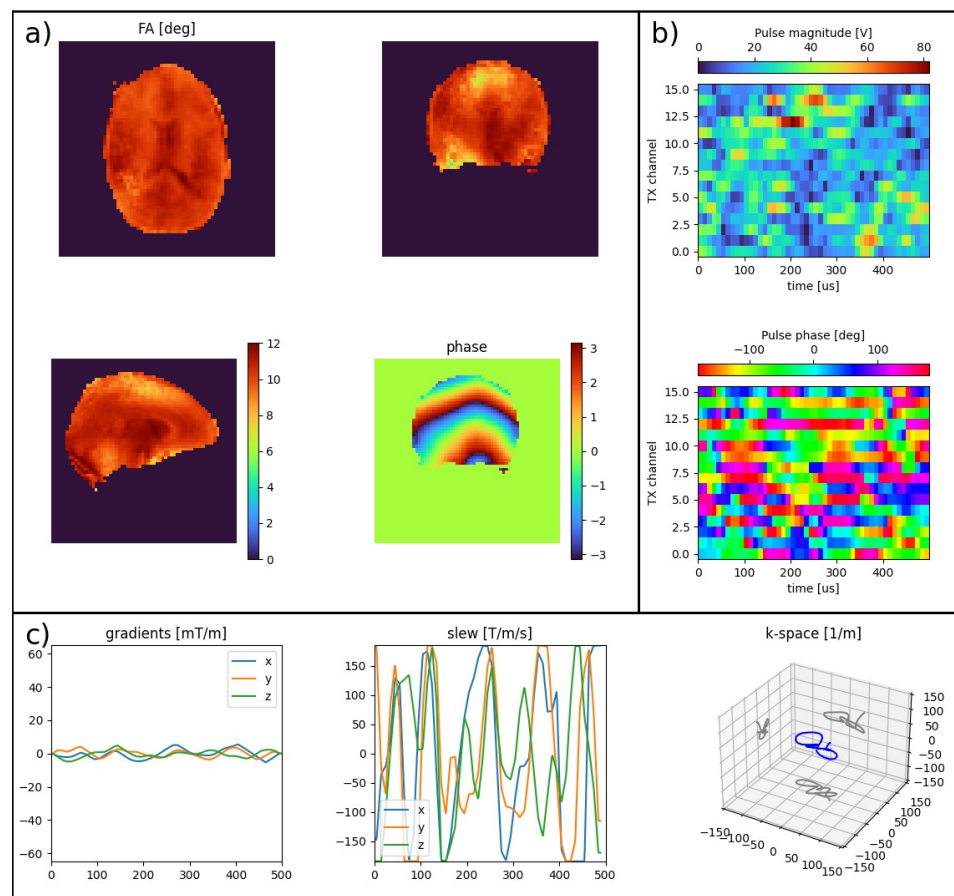
- 9.4T (400 MHz) whole-body magnet
- 16 Tx channels
- Whole-body gradient coil
- In-house RF coil development



THE FastPtx OPTIMIZATION TOOLBOX

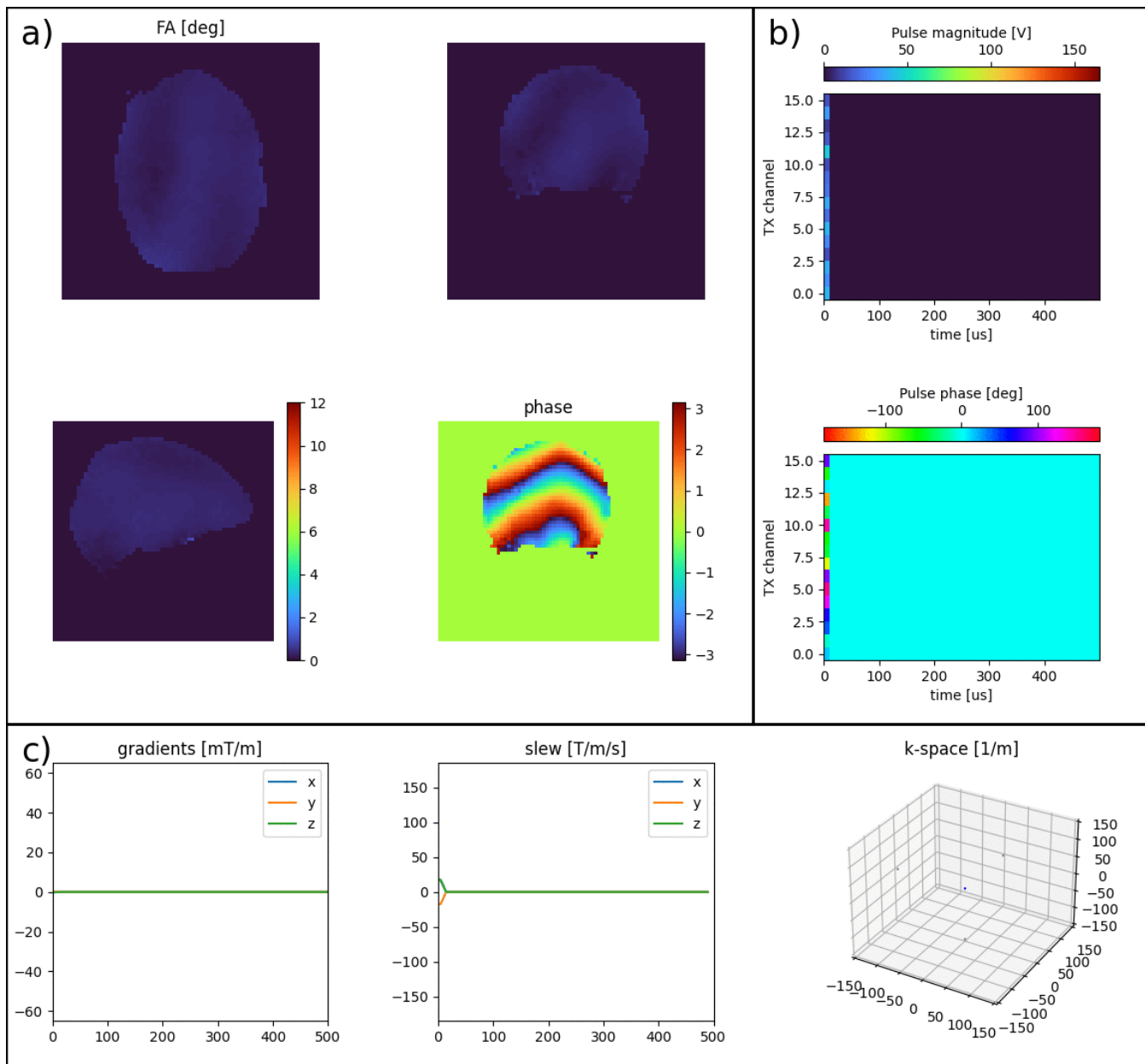
- For each time point: 16 complex RF channels + 3 gradient channels = 35 parameters

- Python → PyTorch
- automatic differentiation
→ efficient
- Supports STA and Bloch simulations
- Can run on CPU or GPU
- Supports individual or universal pulses



FastPtx

Freedom to change all parameters



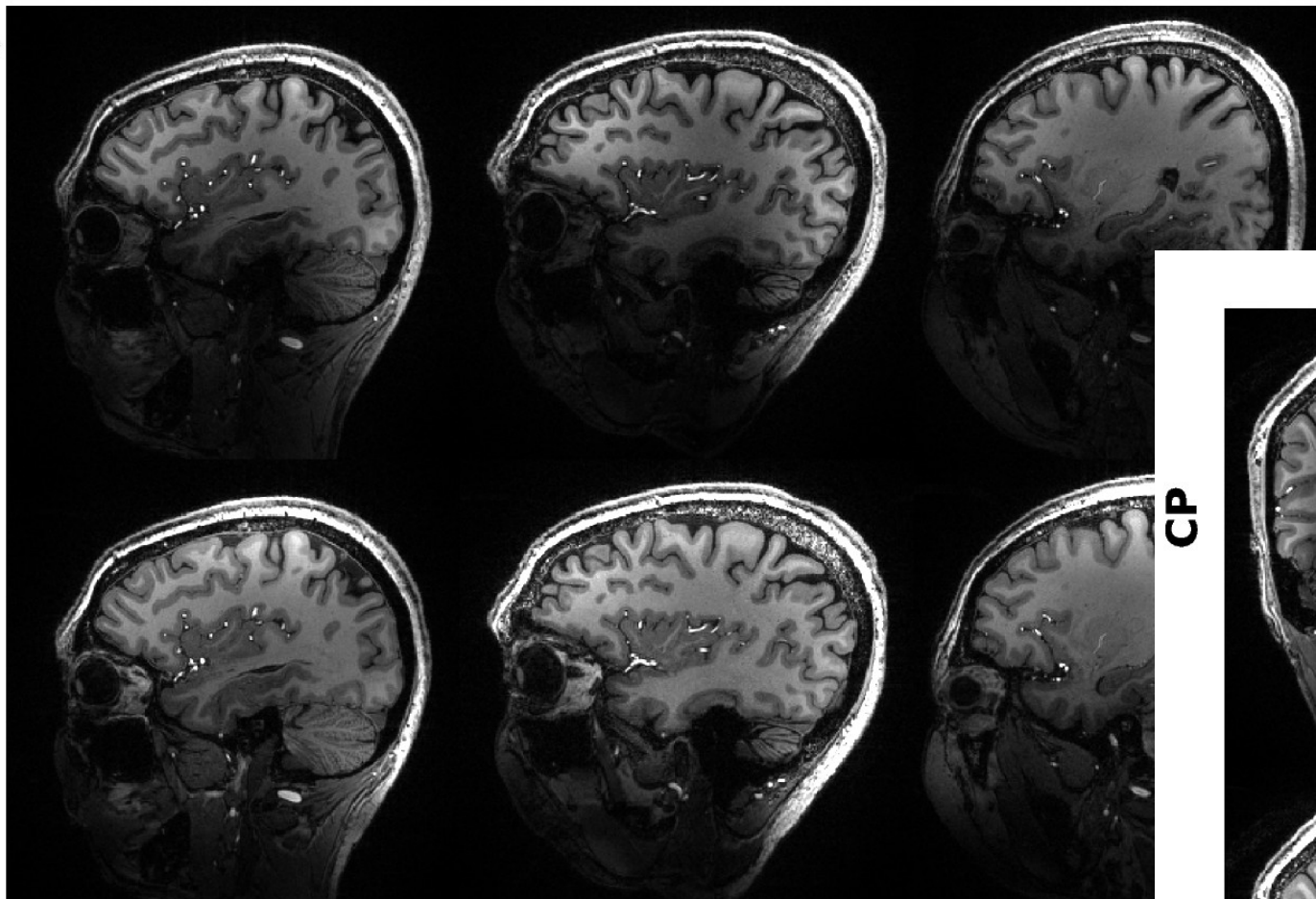
- **Pulse duration: 520 μ s**
=> 1820 parameters
- **Hardware:**
AMD Threadripper 3995WX
(64 cores)
NVIDIA RTX A6000
(48 GiB VRAM)
- **Computing Duration:**
67 seconds
- **This also runs in Colab at reasonable rates**

CP

S1

S2

S3



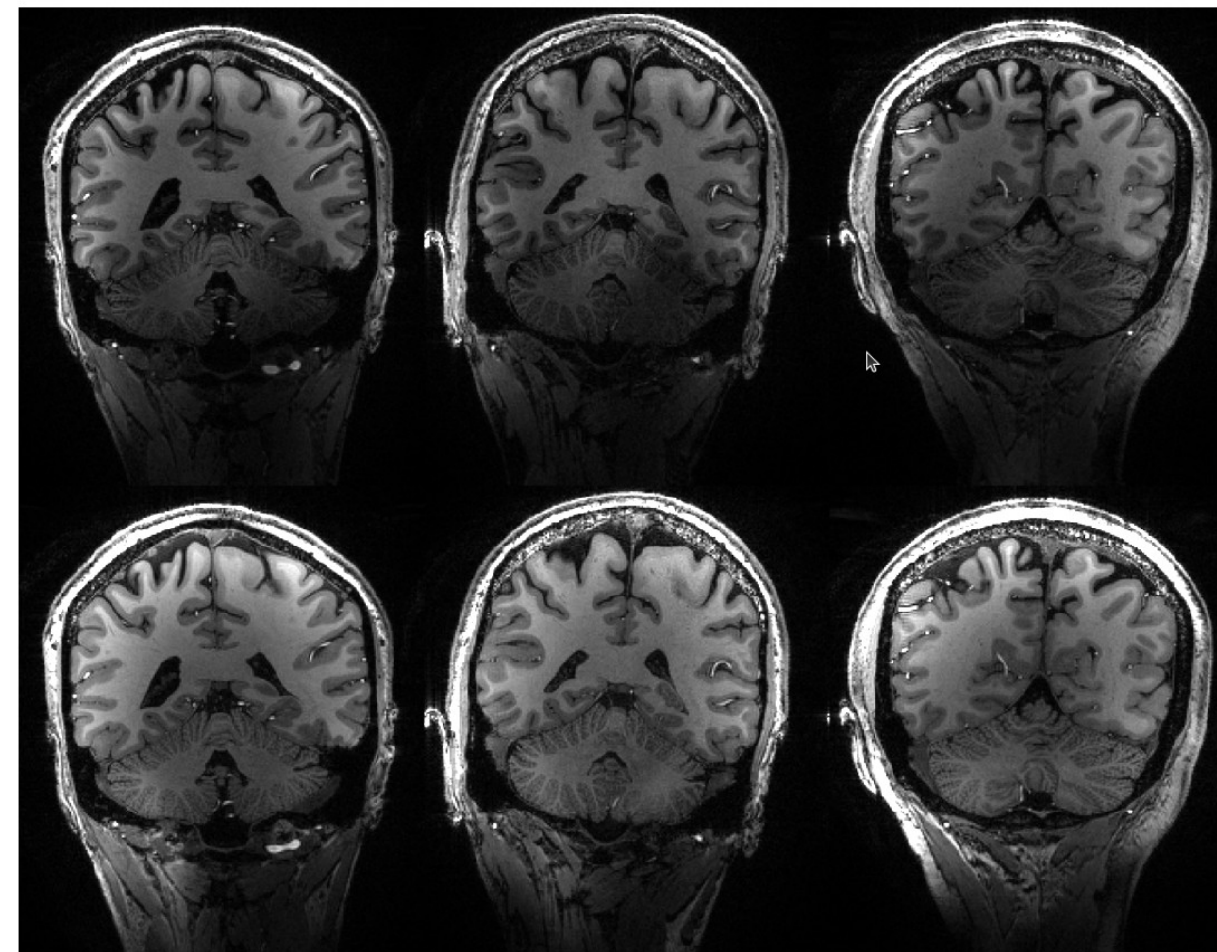
TP

CP

S1

S2

S3



TP

- Toolbox and example data publicly available
- Bosch & Scheffler Magn Reson Mater Phy 2024.
<https://doi.org/10.1007/s10334-023-01134-7>
- github.com/dabosch/FastPtx



https://colab.research.google.com/github/dabosch/FastPtx/blob/main/Colab_Demo.ipynb

